Enhancing 5G Architecture Comprehension with Large Language Models and Multimodal Retrieval-Augmented Generation

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Introduction

- Modern telecom technologies (O-RAN, MIMO, 5G NR, etc.) come with extremely complex specs, often thousands of pages of documentation
- Expanding expertise in this space leads to better, more secure, and higher-performing 5G (and future) infrastructure for billions of users
- This project builds an LLM-based learning tool that uses a Retrieval Augmented Generation (RAG) pipeline to extract and communicate information from 5G specifications
- Core goal: enhance multimodal RAG capabilities and optimize AI as an information retrieval and learning tool by testing diverse RAG methods

Methods

A

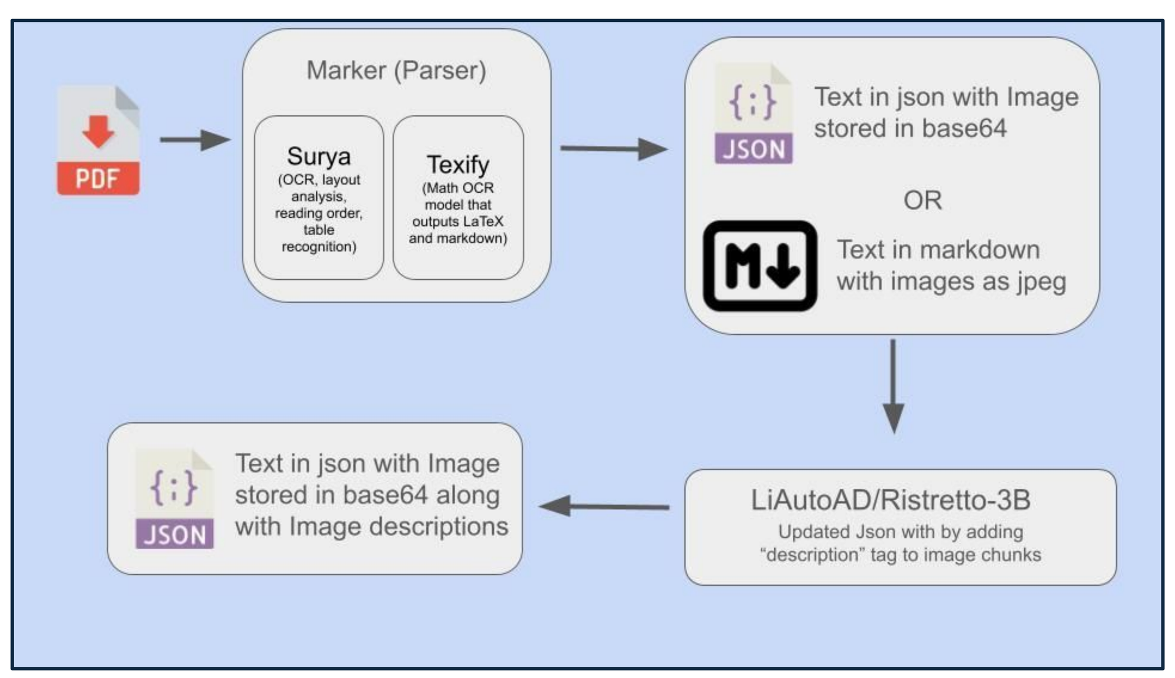


Figure A:

Preprocessing Pipeline
- PDFs are unstructured, so they're first converted to JSON using as computer vision based parser to split documents into chunks based on visual layout
- A lightweight vision-language model, Ristretto-3B, then scans chunks containing images and inserts a text description of each image into the chunk

JSON Database Output
- Each chunk includes both image descriptions (for text-only embedding) and original images (for multimodal embedding), supporting both RAG approaches

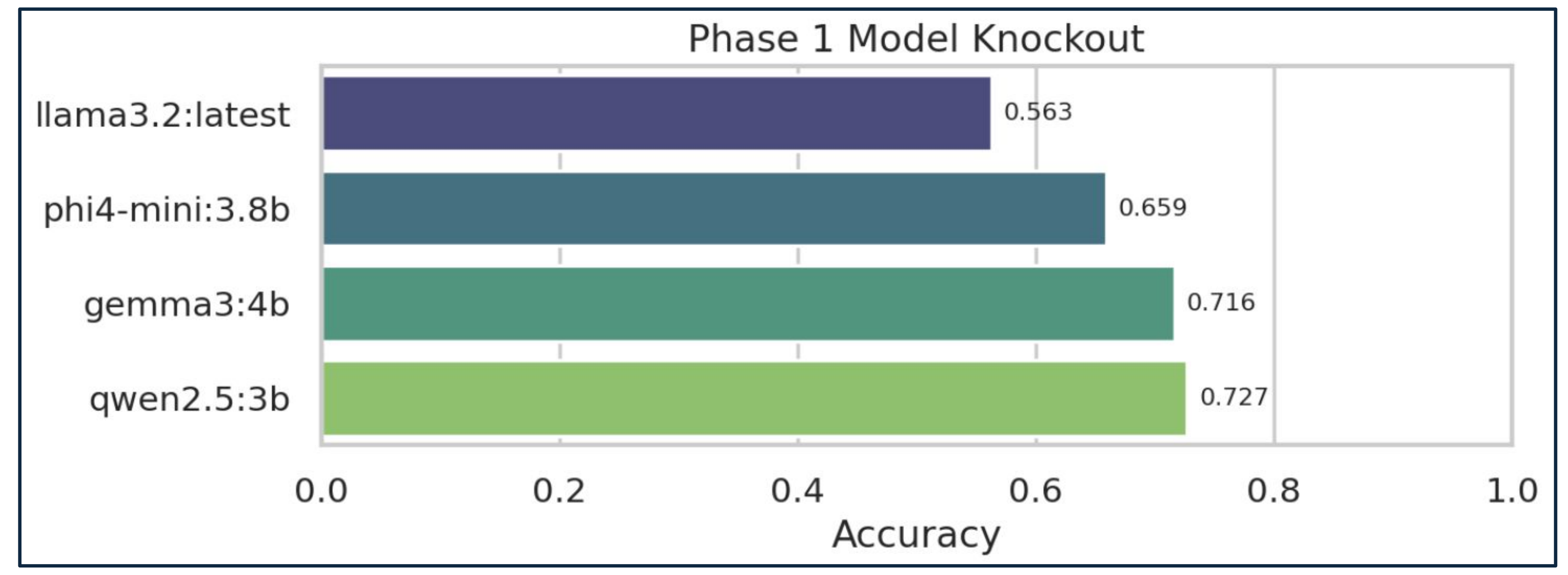
Two Embedding Methods
- Unified Embedding uses **openai-clip-vit-base-patch16** to embed both images and text together
- Grounded Embedding uses **nomic-embed-text,** a text-only model that relies on Ristretto-3B's image descriptions

RAG Hyperparameters
- k-value: number of chunks retrieved during similarity search
- Foundational LLM : base language model used (e.g., Gemma, LLaMA, Phi, Qwen)
- α (alpha) : balances vector embedding vs. keyword search; α = 1 is fully embedding-based, α = 0 is fully keyword-based

Evalulation
- Pipelines were evaluated using ORAN-Bench-13K [12], a multiple-choice benchmark dataset based on O-RAN documentation
- Each pipeline variation took the benchmark test, with its score used as the measure of quality

Results

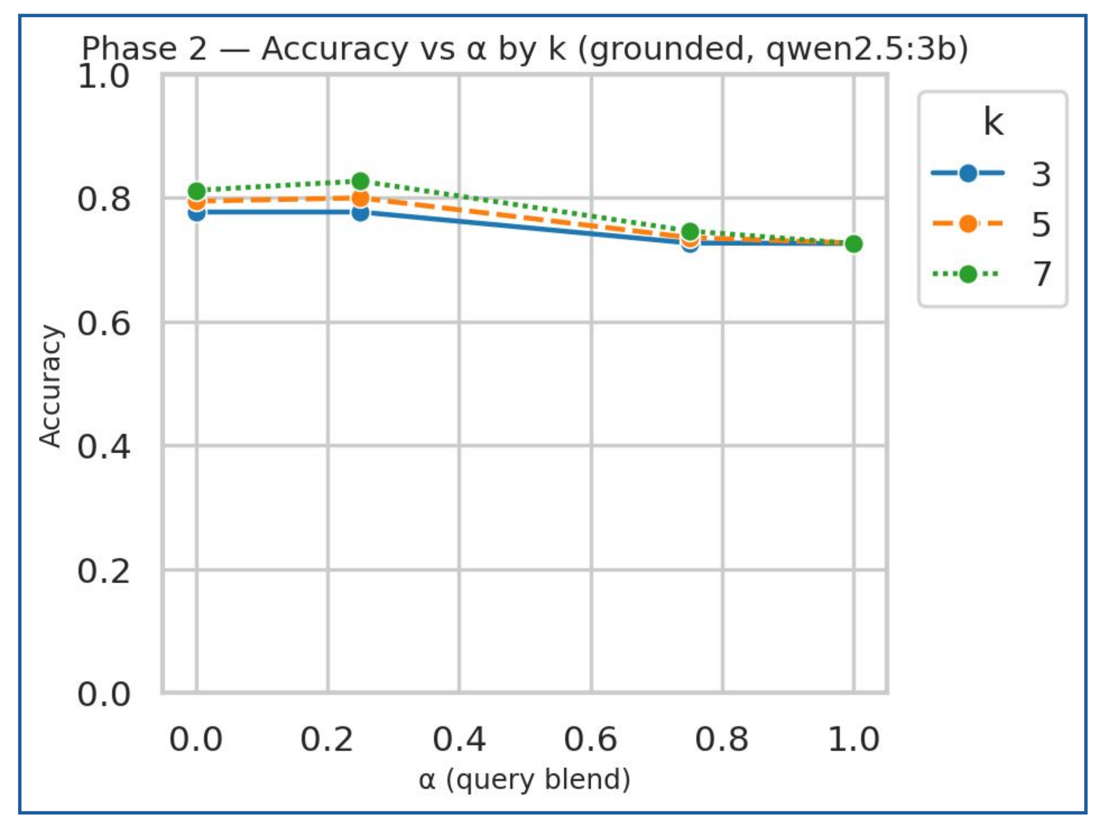
B



Phase 1

- The models were kept within the 3-4 billion parameter range such that they could be judged fairly.
- Based on their initial training, Qwen and Gemma showed the highest results on the dataset and act as our baseline control.

C



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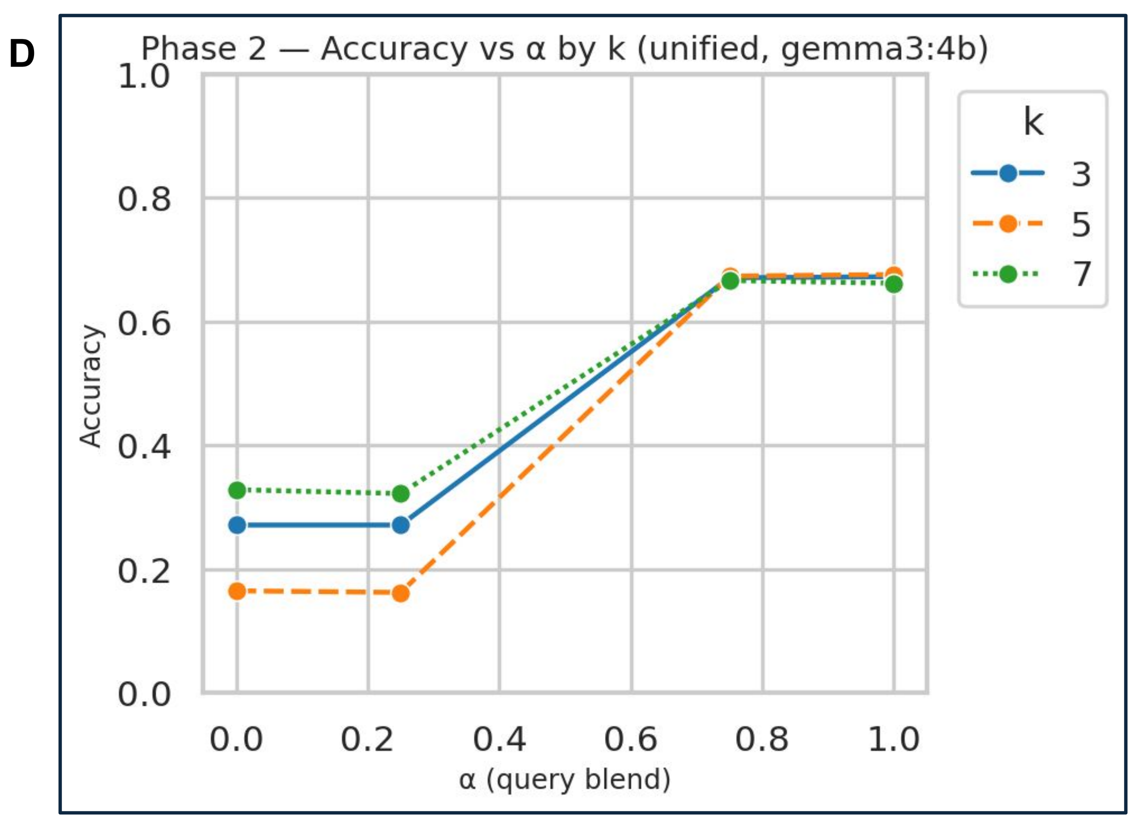
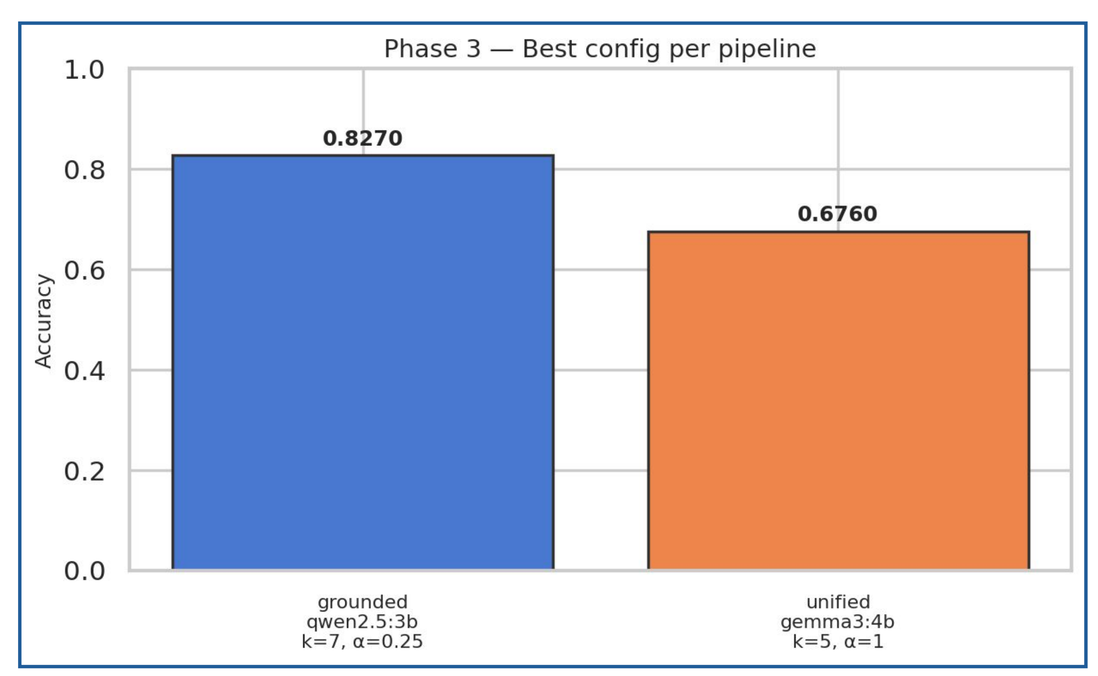


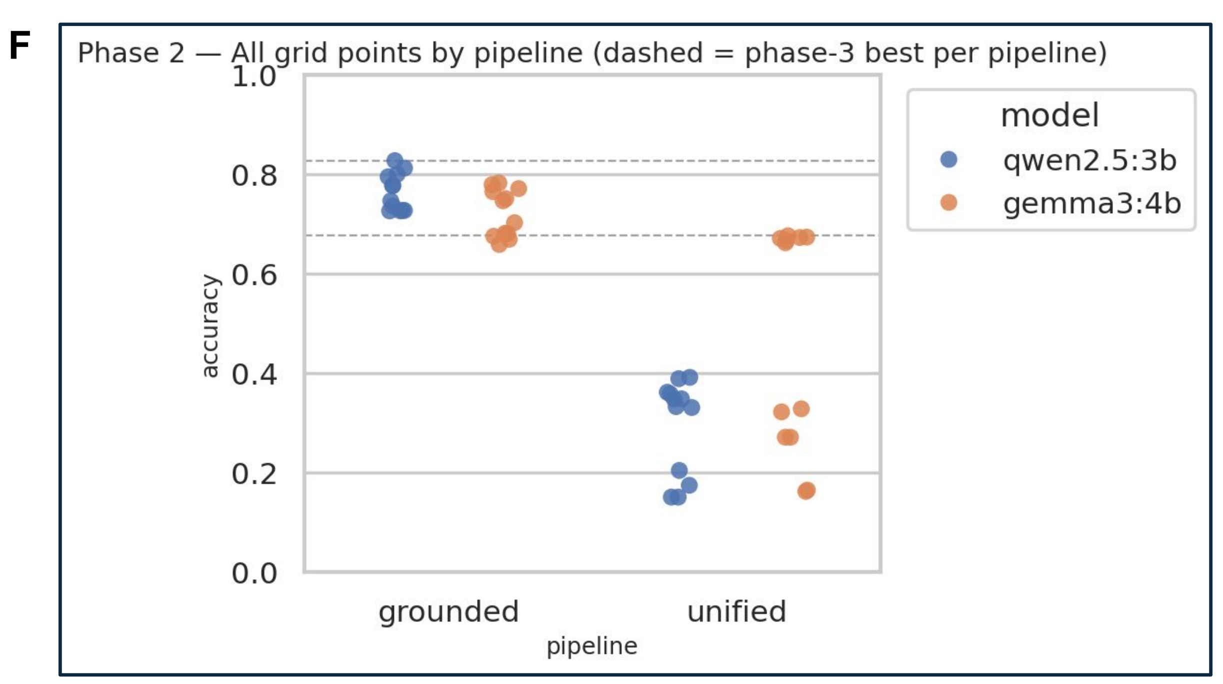
Phase 2

- The results from Figure C show that grounded embedding peaks at 0.82 accuracy, with accuracy falling off as K decreases and peaking at 0.25 with alpha
- Gemma actually performs worse than the control with 0.6760 but mostly maintains it as alpha increases.

E



F



Phase 3

- Grounding drastically increased the accuracy by about 10 percent.
- Grounded embedding is the superior approach for structuring a lightweight RAG pipeline.

Conclusion/Future Work

Key Findings

- Grounded embedding outperformed unified embedding overall, due to the dataset being text-heavy
- When most retrieved chunks are text, unified embedding's cross-modal comparisons introduce noise, retrieving incorrect context and hurting LLM accuracy

Alpha (α) Tuning Insights

- Grounded pipeline performed 10% better at α = 0.25 than at α = 1.0 showing BM25 keyword search vastly improved accuracy
- However, removing vector embeddings entirely (α = 0) caused a slight accuracy drop, suggesting they still contribute a meaningful signal
- Optimal α appears query-dependent, not universal

Future Work

- The two embedding models showed opposite behavior as α increased, meaning embedding type and query blending strategy are tightly coupled and changing one without retuning the other degrades performance
- These findings point toward an adaptive RAG system that dynamically adjusts α based on query characteristics as a promising next step

References

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